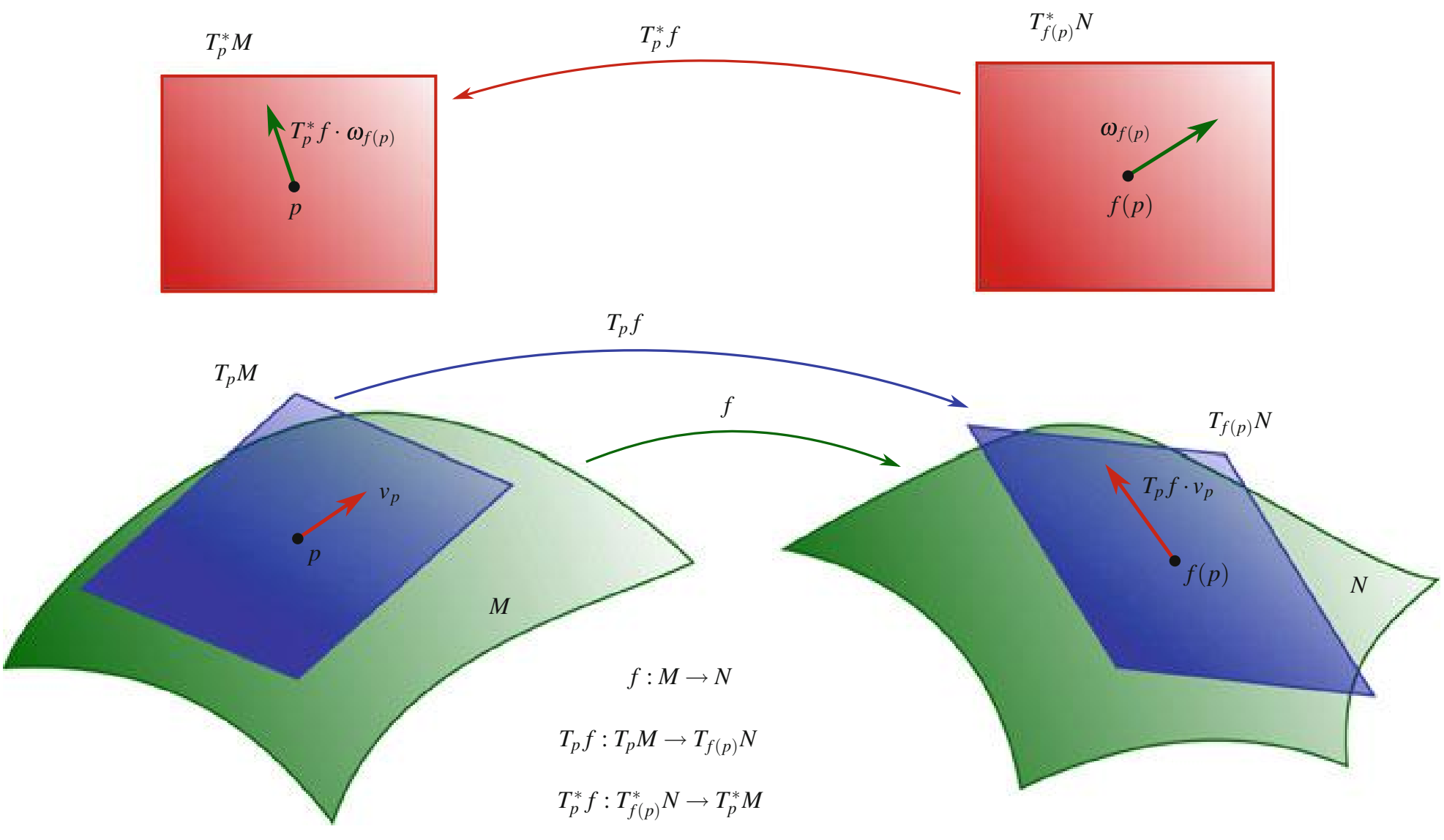

T_p^*M
T_p^*f
$T_{f(p)}^*N$
$T_p^*f \cdot \omega_{f(p)}$
p
$\omega_{f(p)}$
$f(p)$
T_pf
T_pM
f
$T_{f(p)}N$
v_p
p
M
$T_pf \cdot v_p$
$f(p)$
N
$f : M \to N$
$T_pf : T_pM \to T_{f(p)}N$
$T_p^*f : T_{f(p)}^*N \to T_p^*M$